Activity

Scenario - Based - Activity

1. A CCTV image contains salt-and-pepper noise due to transmission errors.

Technique: Median filter.

The median filter is used to remove salt-and-pepper noise, which appears as random black and white dots in an image. It replaces each pixel with the median value of its neighboring pixels. It is suitable for improving the quality of CCTV images.

2. A satellite image is corrupted by Gaussian noise from sensors.

Technique: Gaussian filter.

The Gaussian filter is used to reduce Gaussian noise, caused during image capture. It smooths the image by calculating a weighted average of neighboring pixels using a Gaussian function.

3. Medical x-ray with noise and fine edges.

Technique: Bilateral filter.

Removes noise while preserving important edges and medical details.

4. Scanned document with black and white dots.

Technique: Median filter.

The Median filter removes scattered

black and white dots from the document. It improves text readability while preserving character boundaries.

5. A low-light image shows heavy grain without clear boundaries.
Technique: Non-local means filter.
This filter reduces heavy grain and random noise in low-light images. It also preserves useful textures and image details.

6. A foggy road image has very low contrast.
Technique: Histogram Equalization.
Histogram equalization improves the contrast of the foggy image. It makes vehicles, road markings, and other objects more distinguishable.

7. An underwater image appears dull and washed out.
Technique: Contrast Stretching.
Contrast stretching expands the narrow intensity range of the image. It improves the visibility of underwater objects and details.

8. A chest x-ray shows poor visibility of internal structures.
Technique: CLAHE (contrast limited Adaptive Histogram Equalization)
CLAHE enhances contrast in different regions of the x-ray image. It helps

Make internal structures more distin
without excessively increasing contrast

9. A night-time traffic image lacks
intensity variation.
Technique: Histogram equalization
This technique redistributes pixel
intensity values to improve contra
It helps make vehicles, pedestri
and road signs more visible.

10. A grayscale image uses only a r
intensity range.
Technique: Contrast stretching.
Contrast stretching expands the r
intensity range across a wider ran
of gray levels. This improves th
overall contrast and visibility of
image details.